CLINICAL · LARYNX & AIRWAY

Chronic laryngitis, reflux and specific infections

A voice that has been hoarse for months, in a smoker who clears his throat all day, is the classic picture of chronic laryngitis. Lee warns that the term is not a diagnosis but a description: behind it lie reflux, smoking, voice misuse, infection, allergy and, occasionally, systemic disease, and on the vocal folds some of these can look exactly like cancer. Find the cause, and biopsy anything that does not behave.

AT A GLANCE

- Smoking, reflux and voice misuse dominate.
- Laryngopharyngeal reflux: posterior oedema; twice-daily PPI for months.
- Tuberculosis favours the posterior larynx; lupus the anterior.
- Biopsy persistent lesions.

1 CHAPTER ONE Overview

Prolonged laryngeal inflammation.

1.1 Causes

- **Irritants:** smoking (the main cause), alcohol, dust and fumes in mines and industry, polluted air.
- **Laryngopharyngeal reflux.**
- **Voice misuse:** prolonged loud or forced voice, habitual throat clearing.
- **Infection nearby or below:** chronic sinusitis with postnasal drip, dental and tonsillar sepsis, chronic bronchitis and bronchiectasis.
- **Incomplete resolution** of acute laryngitis.
- **Allergy** as part of the unified airway.
- **Specific infections:** tuberculosis, syphilis, leprosy, scleroma, fungi, and MRSA.
- **Systemic disease:** sarcoidosis, rheumatoid arthritis, granulomatosis with polyangiitis, relapsing polychondritis, amyloidosis, pemphigus and pemphigoid.
- **Radiotherapy.**

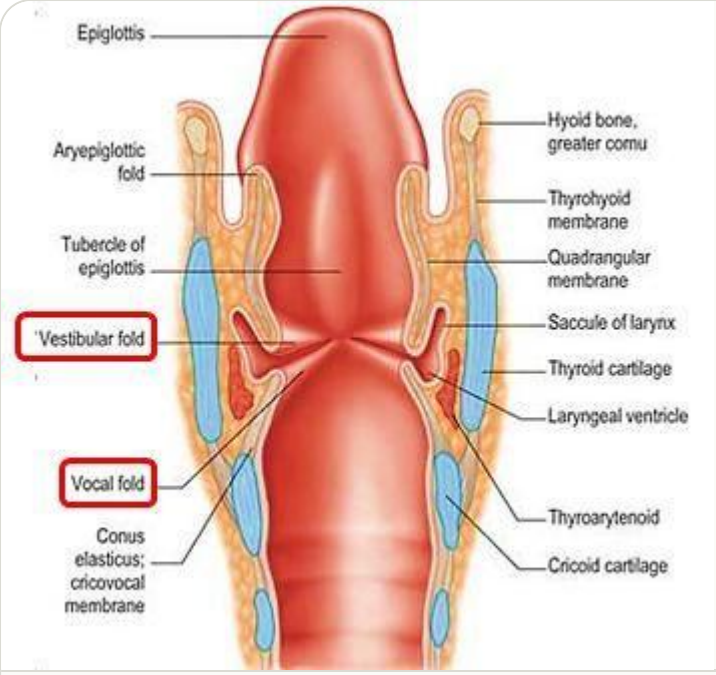


FIGURE 1 Coronal section of the larynx: vestibular and vocal folds, ventricle and saccule.
Anatomy atlas · ENT Steps p. 238

1.2 Approach

Stroboscopic or high-quality endoscopic examination of the larynx; treat reflux, irritants and voice use; biopsy any lesion that persists, is asymmetric, ulcerated or exophytic, or occurs in a smoker, because tuberculosis, granulomas and carcinoma can look alike.



FIGURE 2 Flexible laryngoscopy in clinic.
Dhingra, fig. 62.6

CHAPTER TWO

2 Chronic non-specific laryngitis

Hyperaemic and hypertrophic.

2.1 Chronic hyperaemic laryngitis

A diffuse, symmetrical inflammation of the whole larynx. Hoarseness is the commonest complaint: the voice tires and fails by evening. Constant hawking and clearing of the throat, dryness and tickling, discomfort and a dry cough. The folds look dull red and rounded with flecks of sticky mucus on them and between the arytenoids.

2.2 Chronic hypertrophic (hyperplastic) laryngitis

Mostly men of thirty to fifty. Diffuse thickening that begins at the glottis and spreads to the false folds, epiglottis and subglottis: the respiratory epithelium changes to squamous, the vocal fold epithelium becomes hyperplastic and keratinised, glands first enlarge then atrophy. The mucosa is dusky red and thick,

the folds red, rounded and later bulky and nodular, the false folds swollen (they may be mistaken for prolapse of the ventricle), and fold movement may become sluggish. It can be localised, as vocal nodules, polyps, Reinke's oedema and contact ulcers (described with benign lesions).

2.3 Treatment

1. Treat infection in the sinuses, teeth, tonsils and chest.
2. Stop smoking and alcohol; avoid dust and fumes.
3. Voice rest for weeks, then voice therapy.
4. Steam inhalations and expectorants.
5. Treat reflux.
6. *Stripping* of the hyperplastic mucosa under the microscope in selected cases, one fold at a time and preserving the vocal ligament, with histology to exclude dysplasia.

3 CHAPTER THREE Laryngopharyngeal reflux

Acid at the top.

3.1 Features

Hoarseness, globus, chronic cough, throat clearing, excess mucus and dysphagia; only a quarter to a third have heartburn, because the larynx is injured by small amounts of acid and pepsin that the oesophagus tolerates. Symptoms are worse after late meals, alcohol and exertion; a bitter taste on waking suggests nocturnal reflux.

3.2 Signs

Oedema and redness of the posterior larynx and interarytenoid area, *subglottic oedema* (pseudosulcus), diffuse vocal fold oedema, thick endolaryngeal mucus, and granulomas or contact ulcers.

3.3 Diagnosis

Usually clinical, confirmed by response to treatment. Combined multichannel *pH and impedance* monitoring (detecting acid and non-acid reflux) is the most accurate test; oesophagoscopy excludes Barrett oesophagus.

3.4 Treatment

Diet and behaviour changes (no late meals, less fat, coffee, chocolate and alcohol, weight loss, head of the bed raised), a *twice-daily proton pump inhibitor* half an hour before breakfast and dinner for at least two to three months, an H2 blocker or alginate at night, and fundoplication for proven refractory reflux.

4 CHAPTER FOUR Special forms

Pachydermia, Reinke's oedema and sicca.

4.1 Polypoid degeneration (Reinke's oedema)

Bilateral, symmetrical swelling of the whole membranous vocal folds from fluid in Reinke's space, in middle-aged smokers (women more often seek help because their voice drops), with voice misuse, reflux, chronic sinusitis and myxoedema. The voice is low-pitched and rough; the folds are pale, translucent fusiform swellings that flop with breathing. Stop smoking, treat reflux and hypothyroidism, voice therapy; *microlaryngeal decortication* by a lateral incision with aspiration of the gel and trimming of excess epithelium, traditionally one side at a time.

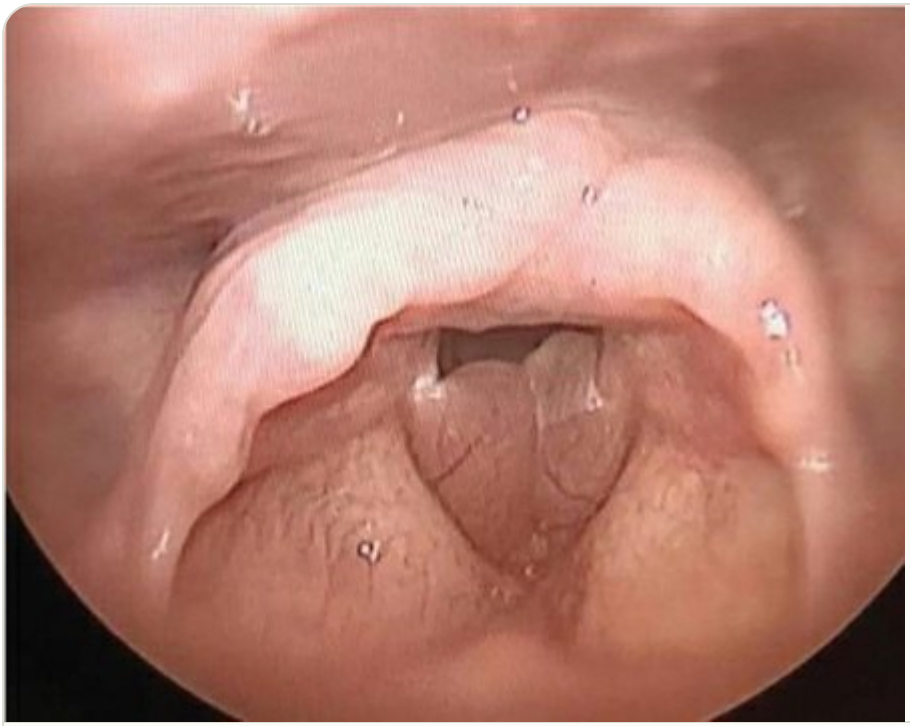


FIGURE 3 Bilateral Reinke's oedema in a smoker.
Dhingra, fig. 61.5

4.2 Pachydermia laryngis

A hypertrophic laryngitis of the posterior larynx: heaped-up red or grey granulation tissue in the interarytenoid area and posterior folds, bilateral and symmetrical, sometimes with contact ulcers on the vocal processes. Associated with alcohol, smoking, forceful speech and reflux. Not premalignant, but biopsy is essential to exclude carcinoma and tuberculosis. Microlaryngoscopic removal, anti-reflux treatment and voice therapy.

4.3 Atrophic laryngitis (laryngitis sicca)

Atrophy of the mucosa with foul crusts, usually in women with atrophic rhinitis and pharyngitis: hoarseness that improves after coughing up a crust, dry cough, and dyspnoea from obstructing crusts, which may reach the trachea. Humidification, sprays of glucose in glycerine, expectorants (ammonium chloride, iodides), and treatment of the nose.

CHAPTER FIVE

5 Specific infections

The great mimics.

5.1 Tuberculosis

Almost always secondary to active pulmonary tuberculosis, mostly in middle-aged men, through infected sputum (bronchogenic) or the bloodstream. It favours the *posterior larynx*: the interarytenoid fold, false folds, true folds and epiglottis in that order.

- **Symptoms:** weakness of the voice (earliest), hoarseness, then severe pain from ulceration radiating to the ears and painful swallowing.
- **Signs:** redness of one fold with impaired adduction (the first sign); a *mamillated* swelling of the interarytenoid area; *mouse-nibbled* ulceration of the fold edge; shallow ulcers and granulations on the arytenoids; *turban epiglottis* (pseudoedema); swollen false folds and aryepiglottic folds; pallor of the surrounding mucosa.
- **Diagnosis:** chest radiograph, sputum, and laryngeal biopsy, essential to exclude carcinoma.
- **Treatment:** antituberculous chemotherapy and voice rest; lesions heal, sometimes with stenosis.

5.2 Lupus

An indolent tuberculous infection accompanying lupus of the nose and pharynx, attacking the *anterior larynx*: the epiglottis first, which may be destroyed, then the aryepiglottic and false folds. Painless and often found incidentally; no lung disease; responds well to antituberculous drugs.

5.3 Syphilis

Now rare; the tertiary gumma forms a smooth swelling that ulcerates and heals with stenosis. Never involved in the primary stage. Diagnosis by biopsy and serology.

5.4 Leprosy

With nasal and skin leprosy: nodular infiltration of the epiglottis, aryepiglottic folds and arytenoids, ulceration, and deformity and stenosis on healing. Dapsone and multidrug therapy.

5.5 Scleroma

Klebsiella rhinoscleromatis, endemic in parts of Africa, the Middle East, Asia and Latin America, usually with nasal disease: a smooth red subglottic swelling causing hoarseness, wheeze and dyspnoea, passing through catarrhal, granulomatous and sclerotic stages to subglottic stenosis. Long courses of tetracycline or ciprofloxacin with steroids; reconstruction for stenosis.

5.6 Fungal

Candidiasis in inhaled steroid users, after antibiotics and in immunosuppression: white plaques on red mucosa that can mimic leukoplakia; systemic antifungals (topical nystatin does not reach), and biopsy if they persist. Blastomycosis and histoplasmosis are rare and mimic carcinoma.

WHERE THE BOOKS DIFFER · WHERE THE BOOKS DIFFER

Dhingra	Lee
Describes chronic laryngitis as an entity, in hyperaemic and hypertrophic forms, with treatment centred on voice rest, steam and removal of irritants.	Advises abandoning "chronic laryngitis" as a diagnosis, replacing it with "prolonged laryngeal inflammation" and a named cause, and gives reflux a central place.

Both insist on biopsy for anything that might be cancer or tuberculosis.

6 CHAPTER SIX Systemic diseases

The larynx as a window.

- **Sarcoidosis:** pale, turban-like submucosal swelling of the supraglottis (epiglottis and aryepiglottic folds); non-caseating granulomas on biopsy; steroids systemically or by injection.
- **Rheumatoid arthritis:** *cricoarytenoid arthritis* (pain, hoarseness, referred otalgia, and when bilateral, stridor from fixed folds) and "bamboo nodules" in the folds; steroids and disease-modifying drugs; tracheostomy for obstruction.
- **Granulomatosis with polyangiitis:** subglottic stenosis, often the first sign; immunosuppression, steroid injection and dilatation.
- **Relapsing polychondritis:** inflammation of the cartilages of the ear, nose, larynx and trachea, with collapse of the airway.
- **Amyloidosis:** waxy yellow submucosal deposits, usually in the false folds and ventricles, with apple-green birefringence on Congo red staining; endoscopic excision and screening for systemic disease.
- **Pemphigus and pemphigoid:** blisters and scarring of the supraglottis.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes of chronic laryngitis,** and why biopsy matters.
- **Laryngopharyngeal reflux:** symptoms, signs and treatment.
- **Reinke's oedema** and **pachydermia laryngis.**
- **Tuberculosis of the larynx:** sites, signs (mouse-nibbled, turban epiglottis, mamillated interarytenoid) and diagnosis; **lupus** compared.
- **Scleroma** and **cricoarytenoid arthritis.**

7.2 Common pitfalls

COMMON PITFALLS

- Treating persistent hoarseness in a smoker with antibiotics instead of laryngoscopy.
- Forgetting the chest radiograph in a patient with posterior laryngeal ulceration.
- Giving a short, once-daily course of proton pump inhibitor for laryngopharyngeal reflux and calling it a failure.

Written for this app from Oxford Textbook of Otolaryngology ch. 47, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 12, Dhingra ch. 58 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · TUMOURS

Supraglottic Tumor

IMAGING FINDINGS

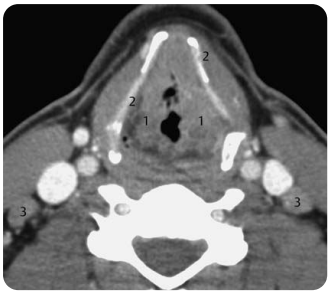
- Mass on the epiglottis, aryepiglottic folds or false cords
- Invasion of pre-epiglottic or paraglottic fat (T3)
- Nodes common, often bilateral

TOP DIFFERENTIALS

- Laryngocele
- Benign mucosal swelling
- Minor salivary tumour

CHECKLIST

- Vallecula and tongue base; pyriform sinus
- Cartilage and extralaryngeal spread



CT with intravenous contrast enhancement, axial. The lesion extends bilaterally (1), highly indicative of a malignancy. The…



Axial CECT shows similar findings with invasion of bilateral paraglottic fat and left AE fold. Tumor bulges through thyroid cartilage notch. Pyriform sinu…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 526.